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CALCULATION OF X5 AND PLOTTING OF 3D GRAPHS FOR VEL WITH NEGATIVE VALUES OF VARIABLES

Two years ago, the Nobel Prize in Economics was awarded to three American economists who significantly improved the understanding of the role of banks in the economy, especially during financial crises, by giving money to banks and the population to restore purchasing activity in the country. Here it is worth recalling at once that in a number of countries, such as Sweden, Switzerland, Denmark and Japan, banks issued some customers with a negative rate, i.e. customers returned an amount less than they received [1, 2, 3].

This article considers the issue of calculating the values of the lower critical volume of the Vel spherical economic shell and constructing 3D drawings when the values of three variables change, i.e. Vel (GDPel) = f(X1, X2, X3, X4, X5). In some cases, when the Vel values had small minima and maxima compared to the last calculated Vel value, then 3D drawings were built from the calculation so that the reader would understand more clearly how these changes occur.

Here, Figure 1 shows a 3D graph of Vel, when the values of the variables were as follows X1= X3= -0.1...-1, X2= -1...-0.1, X4= 1, X5= 1.36...1.41. Here Vel increases by 1.08 times. In the following Figure 2 the depicted 3D Vel graph at variables X1= X3= -1...-0.1, X2=-0.1...-1, X4= 1, X5= 1.41...1.36 decreases by 1.08 times.

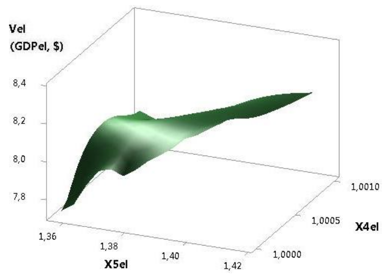

Fig. 1. 3D of Vel (GDPel)

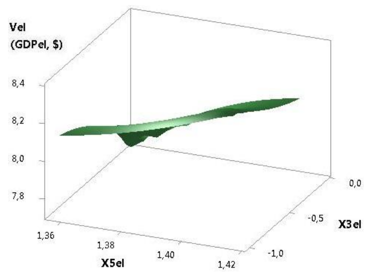

Fig. 2. 3D of Vel (GDPel)

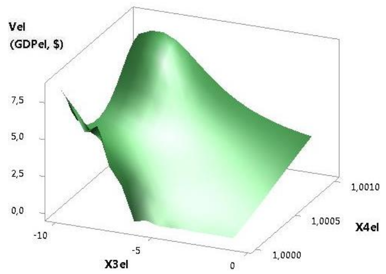

Fig. 3. 3D of Vel (GDPel)

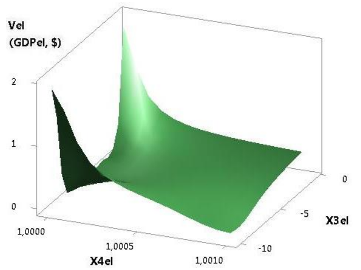

Fig. 4. 3D of Vel (GDPel)

The following two figures 3 and 4 show 3D Vel graphs when the variables were X1= X2=-1...-10, X3=-10...-1, X4= 1, X5= 1.41...0.76 and X1=-1...-10, X2= X3= -10...-1, X4= 1, X5= 0.67...0.67. As you can see, the plotted 3D Vel graph in Fig. 3 is reduced by 3.4 times, and in Fig. 4 it has only two solutions at points 1 and 10.

The plotted 3D Vel graph in Figure 5 increases 3.05 times at variables X1= X2= X3= -1...-10, X4= 1, X5= 0.78...1.36. From the following Figure 6, it can be seen that at variables X1= X2 = -10...-1, X3= -1...-10, X4= 1, X5= 0.76...1.41, Vel increases 3.4 times.

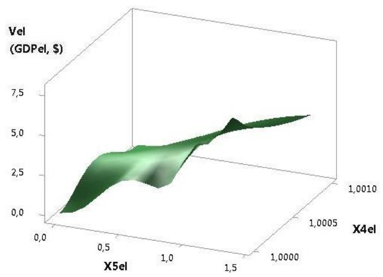

Fig. 5. 3D of Vel (GDPel)

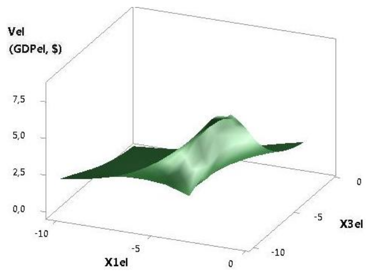

Fig. 6. 3D of Vel (GDPel)

Figures 7 and 8 were plotted at X1= X3= -1...-10, X2= -10...-1, X4= 1, X5= 0.78...1.36 and X1= X3= -10...-1, X2= -1...-10, X4= 1, X5= 1.36...0.78. Here in Fig. 7 The Vel values increase by 3.78 times, and in Figure 8 it decreases by 3.05 times.

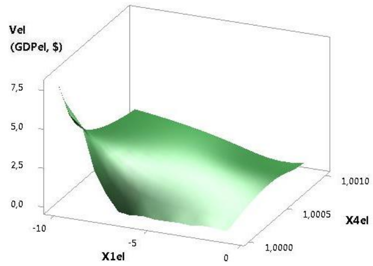

Fig. 7. 3D of Vel (GDPel)

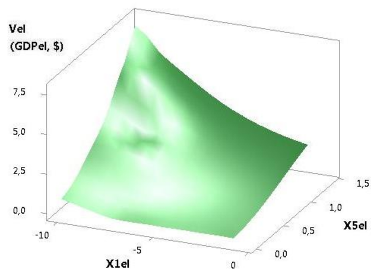

Fig. 8. 3D of Vel (GDPel)

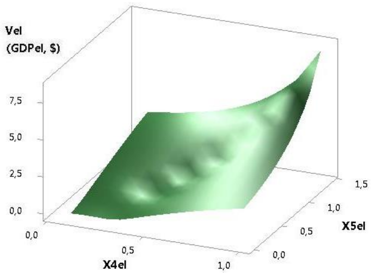

Fig. 9. 3D of Vel (GDPel)

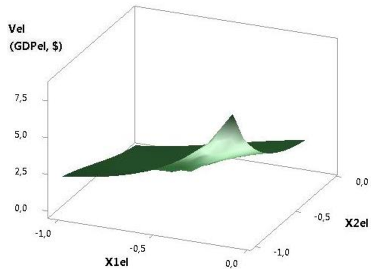

Fig. 10. 3D of Vel (GDPel)

The following Figure 9 was plotted at X1= X2= -0.1...-1, X3= -1...-0.1, X4=1...0.1, X5=1.41...0.43. Here the Vel parameter decreases by 26.72 times. The Vel parameter, plotted in Fig. 10, at X1= -0.1...-1, X2= X3= -1...-0.1, X4=1...0.1, X5=1.41...0.07 decreases 17.94 times.

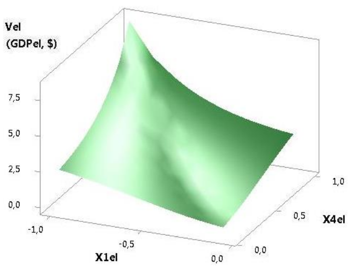

Fig. 11. 3D of Vel (GDPel)

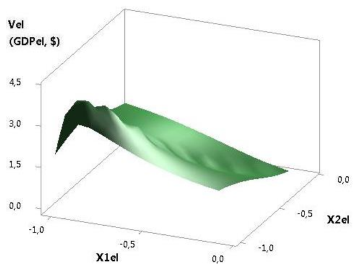

Fig. 12. 3D of Vel (GDPel)

The last figures 11 and 12 were plotted at X1= -1...-0.1, X2= X3= -0.1...-1, X4=1...0.1, X5= 1.41...0.07 and X1= X2= -1...-0.1, X3=-0.1...-1, X4= 1...0.1, X5= 0.67...0.14. In these examples in Figure 11, the Vel variable decreases by a factor of 17.94, and in Figure 12 it has a maximum of 3.98 at point 3.

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